

Introduction

This project focuses on the design and implementation of an amplitude modulation (AM) based wireless audio transmitter and receiver system. The objective was to transmit audio signals wirelessly using modulation principles and recover them at the receiver end.

A transmitter circuit was designed to modulate an audio signal onto a high-frequency carrier, while a receiver circuit was developed to detect and recover the original signal. Experimental testing demonstrated successful wireless transmission of audio over a distance of approximately **4–5 cm**.

Detailed analysis of the system behavior revealed that the signal transfer was primarily dominated by **near-field magnetic coupling (inductive coupling)** rather than efficient far-field electromagnetic radiation. The project provided practical insights into modulation techniques, electromagnetic induction, and real-world challenges in wireless communication systems.

Objectives

The objectives of this project are:

- To understand the working principle of amplitude modulation
- To design and construct an audio transmitter circuit.
- To design and construct a receiver circuit capable of signal detection.
- To experimentally study wireless signal transmission.
- To analyze practical limitations in real circuit implementation.

Theoretical Background

Amplitude Modulation

Amplitude modulation is a technique in which amplitude of a high frequency carrier is changed according to amplitude of message signal

Amplitude modulation has 4 types

- 1.) Double side band full carrier (DSBFC)
- 2.) Double side band suppressed carrier (DSBSC)
- 3.) Single side band (SSB)
- 4.) Vestigial side band (VSB)

This project for sake of simplicity used DSBFC method of amplitude modulation. This method though a bit less efficient but is simple to implement relatively as compared to other methods.

Signal Transfer

Signal transfer between transmitter and receiver occurs due to electromagnetic induction from transmitter coil to receiver coil.

A time-varying magnetic field generated by the transmitter induces voltage in the receiver coil.

Components used

S.No.	Components	Specifications	Quantity
1.)	Resistors	-	Net 11 of different values as shown in circuit diagrams
2.)	Capacitors	Ceramic ones in tank	Net 11 of different values as shown in circuit diagrams
3.)	Inductors	100uH	2
4.)	Transistors	2	BC109(Metal can)
5.)	Amplifier module	LM386 module	2
6.)	Mic	Electret microphone	1
7.)	Speaker	8ohm 0.5W	1
8.)	Diode	OA79	1
9.)	Battery	9V	2

Circuit Design

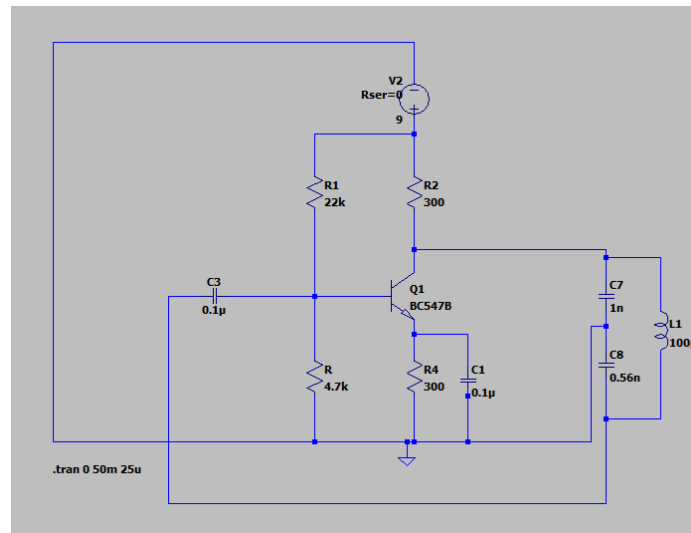
Transmitter Circuit

This part consists of:-

- Colpitts oscillator circuit
- Audio generation and amplification circuit
- Modulation circuit
- Transmitting coil

Note :- For modulation LM386 audio amplification module is used for reliability

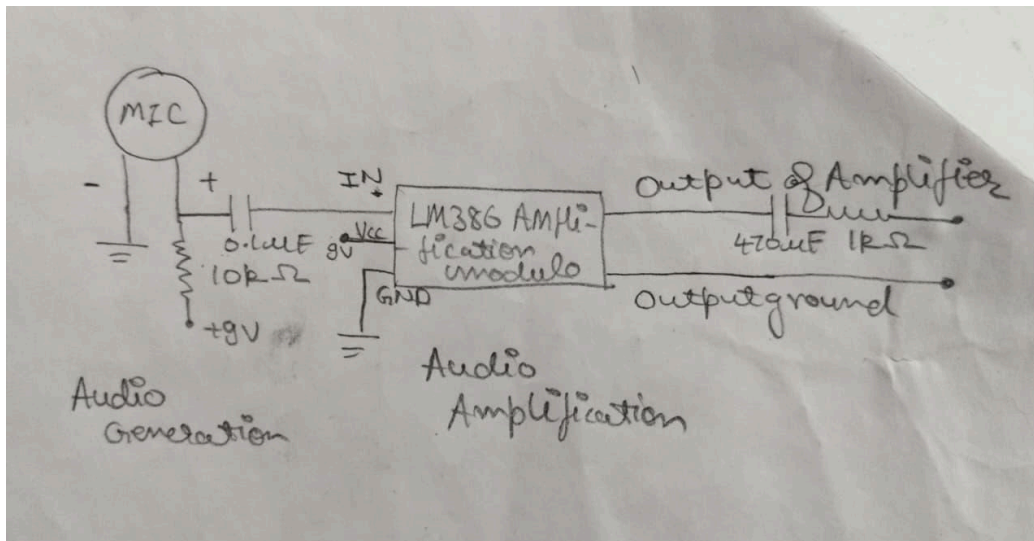
Colpitts oscillator



Circuit Diagram of Colpitts oscillator

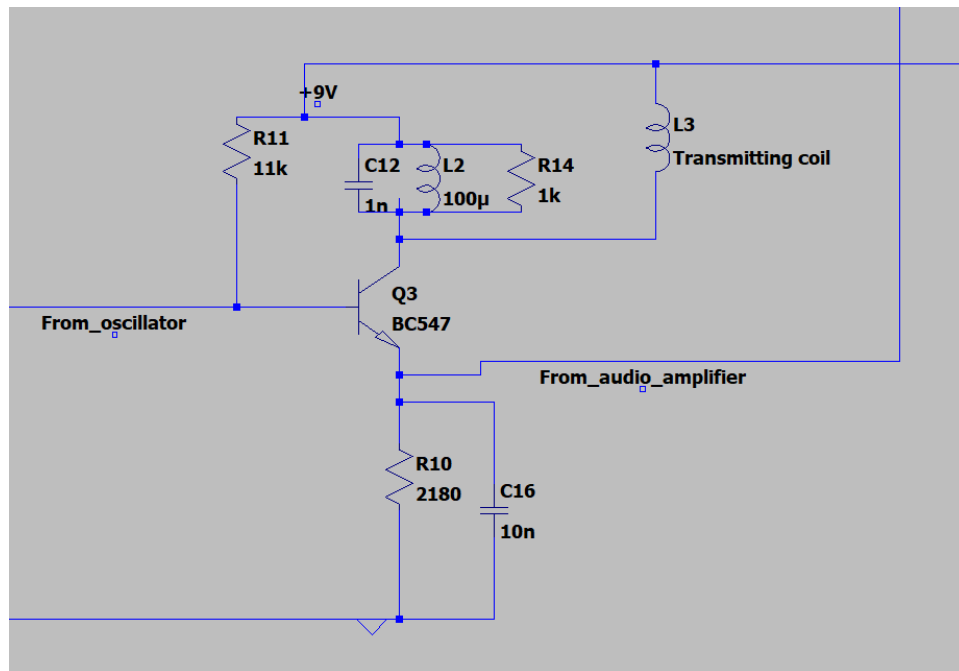
- Oscillating frequency of this oscillator can be found out formula as : $f = 1/(2\pi\sqrt{LC})$, where $C = (C7 \cdot C8)/(C7 + C8)$, and hence $C \sim 0.36\text{nF}$
- Calculating oscillating frequency we obtain 838kHz as oscillating frequency

Audio generation and amplification



Audio Generation and amplification block diagram

Modulator and transmitting coil

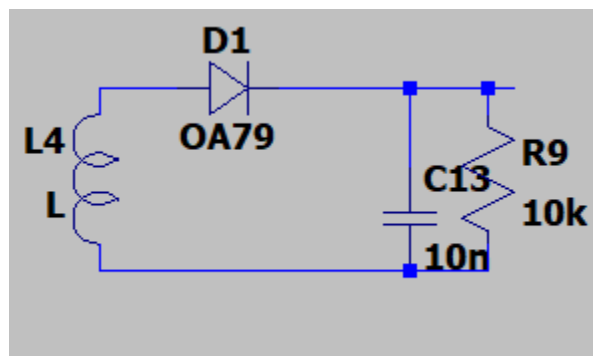


Circuit Diagram of modulator

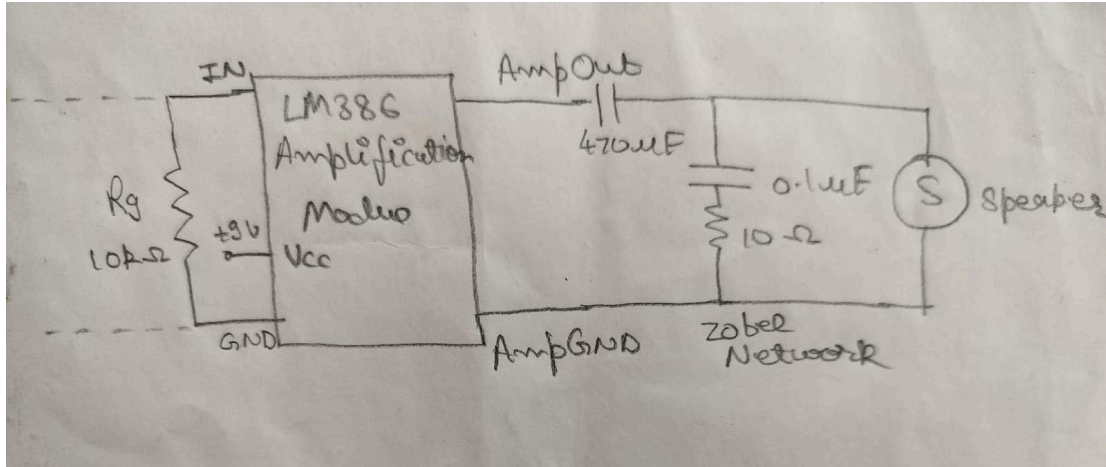
- An amplifier which amplified carrier wave whose gain is determined by current from audio wave (emitter current)
- Transmitting coil is essentially a 22AWG wire of 20 turns of diameter around 7.5cm

Receiver Circuit

- Receiver circuit consists of a receiver coil mirroring the transmitting coil, a peak rider circuit as demodulator and LM386 audio amplification module and a speaker.



Circuit Diagram of peak rider circuit



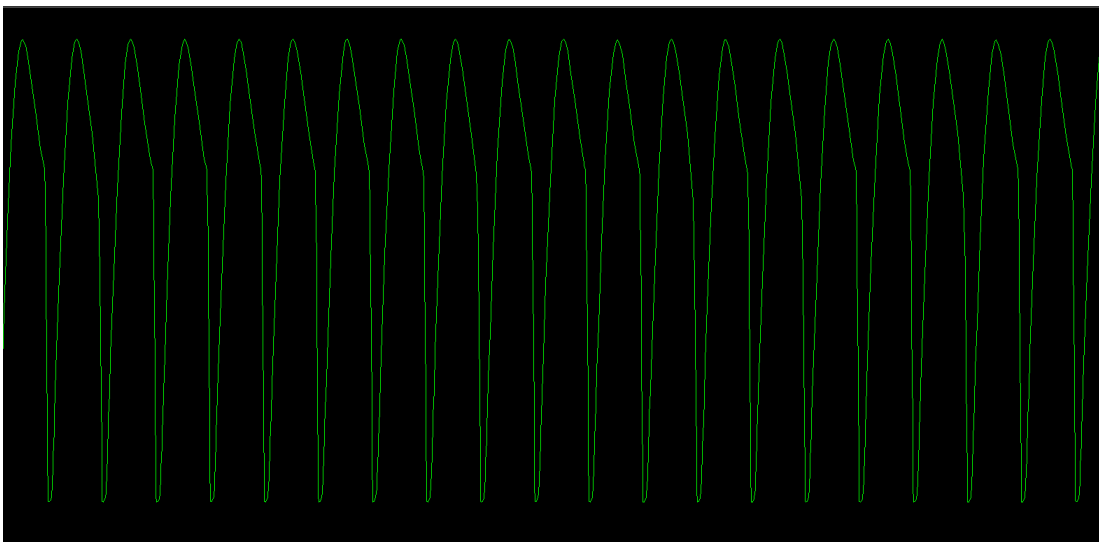
Block diagram of audio amplification and playing on speaker

Procedure

- Circuit was implemented on breadboard
- While testing reception coil was probed to DSO and a 1kHz sine wave was put as input
- Receiver coil was placed near and in alignment with transmitting coil
- Signal Quality and transmission range were observed

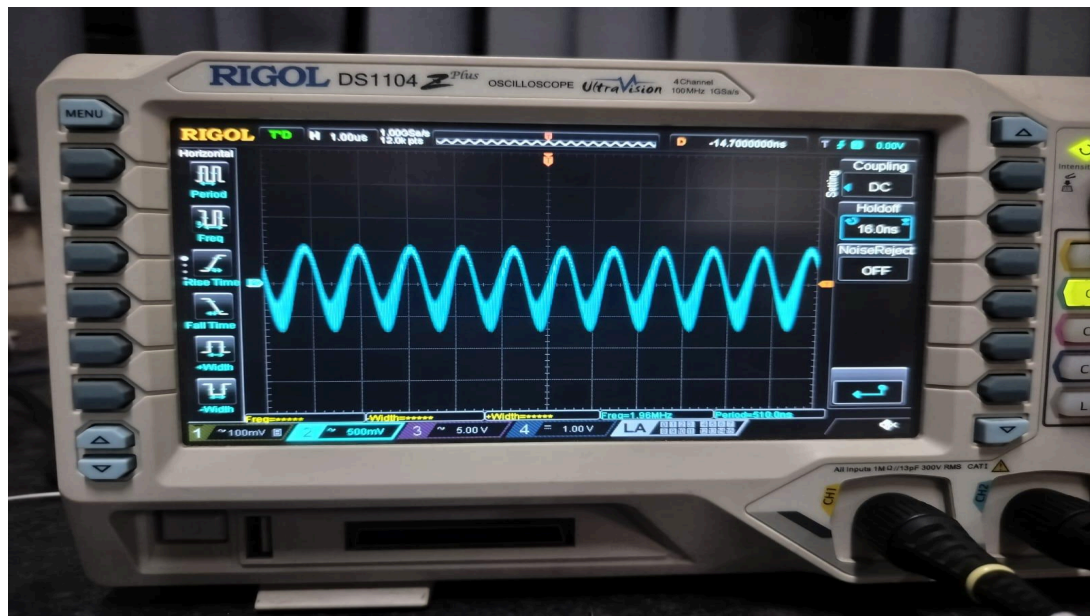
Observations

Carrier wave



Approximate observed carrier wave

Note :- The carrier wave is not a sine wave due to high gain of amplifier , this was verified by removing the emitter bypass capacitor and observing a sine wave on DSO

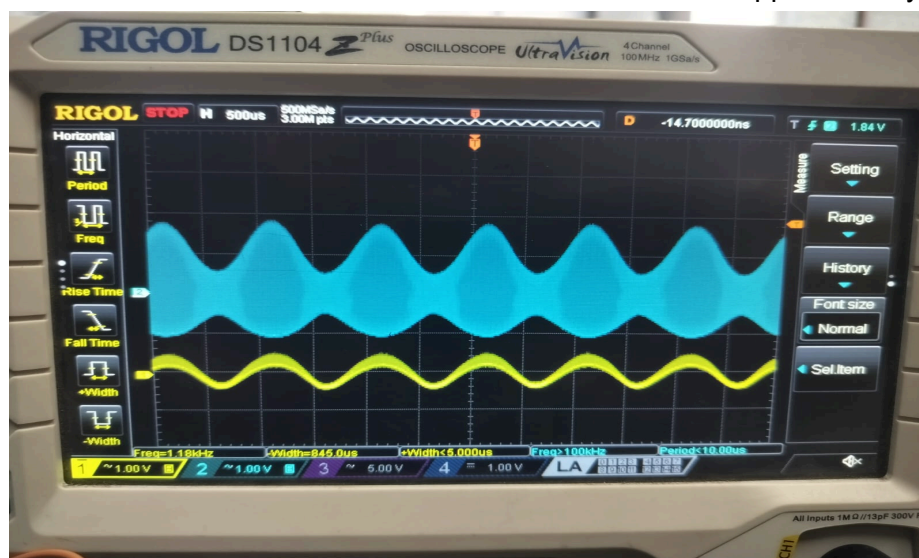


Sine wave without emitter bypass capacitor observed on DSO

Also note that carrier wave attached is not from DSO it is the approximate shape observed on DSO but is from simulated oscillator.

Audio Modulation

Modulation index observed on an unloaded modulator was approximately 0.6



Probed modulator output (blue) and audio input(yellow)

- There are distortions in the input audio wave of modulator cause of loading effect of modulator.

Modulation index dropped to 0.45 on connecting the transmitting coil due to change in impedance and loading



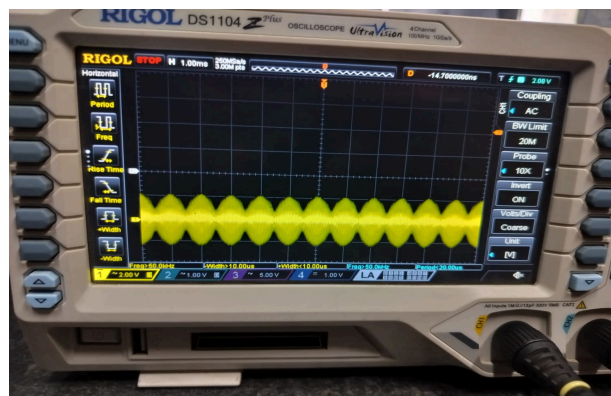
Probed modulator output(blue) and audio input(yellow)

Note:- The lower band of wave isn't modulated properly due to gain and biasing issues ,but it doesn't matter as both bands carry the same information and we can extract the information from the upper envelope.

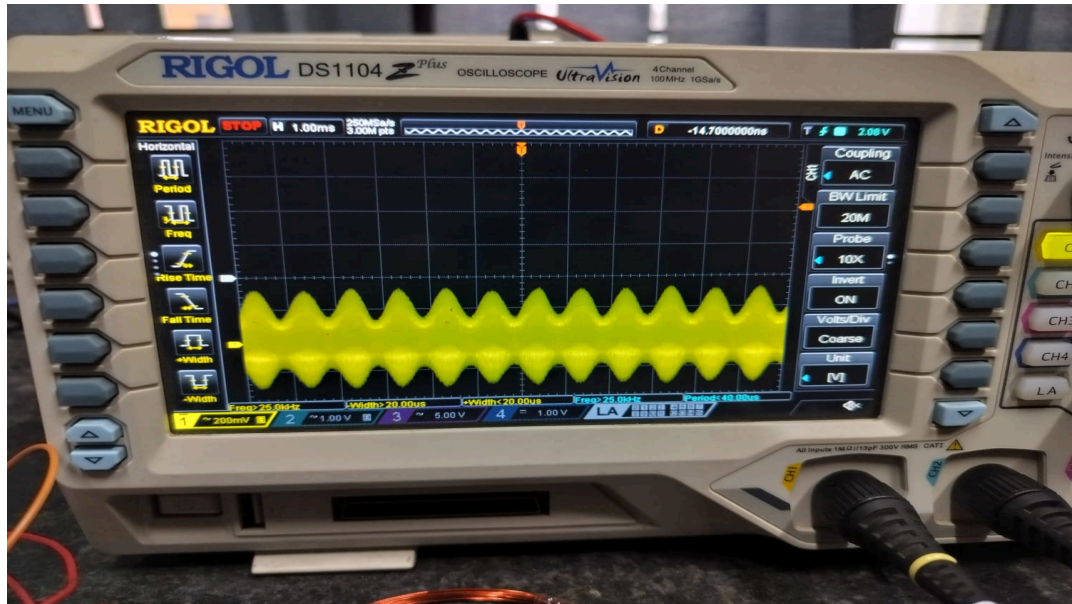
Audio Recovery

Receptor coil without any further circuitry was probed to observe the induced voltage

- It was observed when transmission and reception were kept extremely near(touching each other), due to cross coupling received wave was distorted
- Only when the coils were kept at distance of 2-3cm or more, a less distorted wave was observed across receptor coil

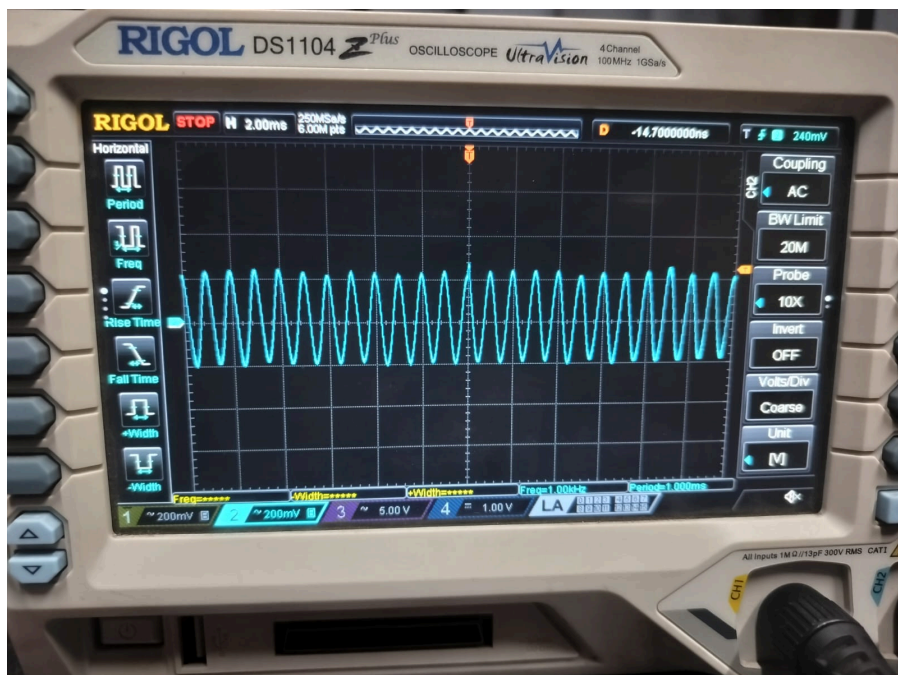


Distorted voltage across receptor coil



Less distorted voltage obtained across the receptor coil

- After demodulation for a input sine wave of 1kHz, this was obtained across resistor of peak rider circuit indicating successful transmission of audio wave



Obtained wave on output of peak rider circuit

Note:- The dominant way of information transfer in this experiment was inductive coupling, rather than EM radio waves
Also note that on amplification and playing on not perfect zobel network there were distortions in this sound which lead to a hissing sound out of speaker

Recorded audio wave from speaker can be heard here:-[Output.mp4](#)

Results

The project successfully demonstrated wireless transmission of audio signals.

However, experimental observations showed that:

- Signal strength decreased rapidly with distance.
- Reliable reception occurred only within approximately 4–5 cm.
- Coil alignment strongly affected signal quality.

These observations indicate that the communication mechanism was dominated by **near-field magnetic coupling** rather than efficient radiated electromagnetic waves.

This is mainly because:

1. No optimized antenna structure was used.
2. Most energy remained confined in the near field.
3. Radiation efficiency of the transmitting structure was low.

Changes

There are some changes made from earlier submitted design to make the project work out

- The tank circuit earlier consisted of (1uF+0.1uF)capacitors + (0.47uH) Inductor.
- One of the problem with this tank was due to large value of capacitors ,it worked on large value of current which the transistor was unable to provide and there was no oscillations observed.
- Another issue was that the 0.47uH inductor was not available as an standard plug in type inductor, it had to be hand wounded ,which is unreliable and very sensitive.
- Hence the tank was changed ,and capacitors were made (1nF+560pF) and (100uH) inductor, now capacitors are light enough to be driven by transistor and also 100uH was available as standard inductor.
- Another problem encountered was of modulation index, initially with the proposed design ,it didn't exceeded 0.2, which was very less



Modulated wave with proposed design(Blue) and audio wave(yellow)

- To change this two changes were made
- First was to remove the base to ground resistor to push in more carrier wave.
- Second was that, to increase emitter to ground resistance to avoid the leakage of audio wave into ground and increase emitter current and hence more gain influence of audio wave
- Another change was shift from transistor based audio amplifier to a amplification module for more reliability, this is the only module used in entire project
- Changes in receiver include removal of filter cause of the clean 1kHz audio wave obtained at output, there felt no need of adding a filter.
- Another change was making peak rider circuit more robust by increasing resistance and decreasing capacitor.

Limitations

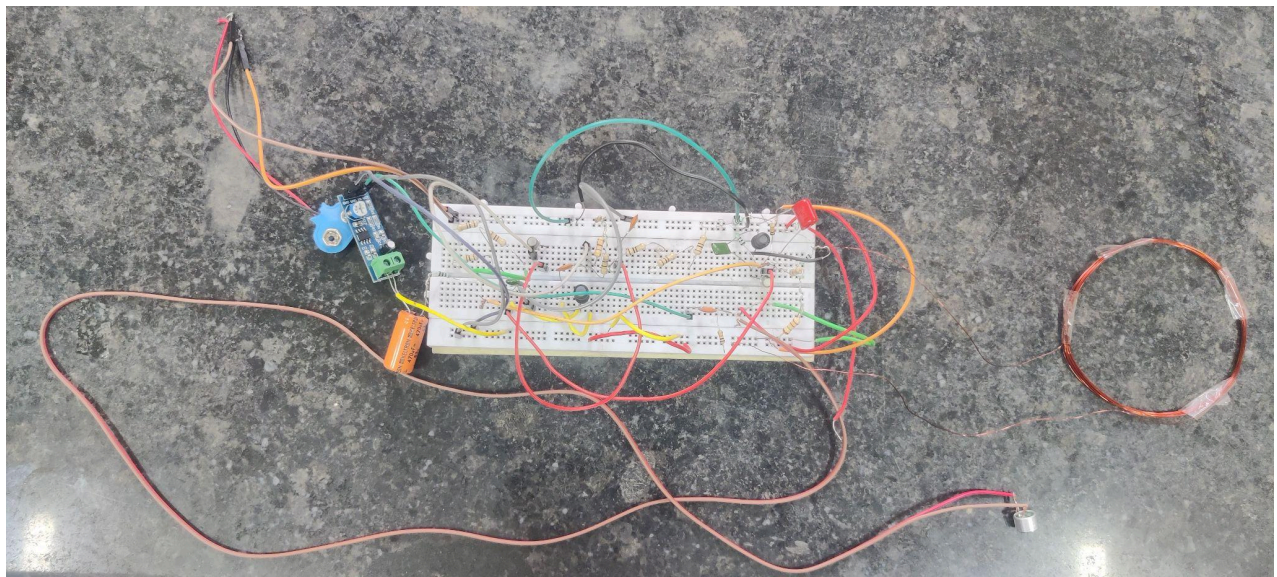
The following limitations were observed:

- Limited transmission range.
- High sensitivity to alignment.
- Low radiation efficiency.
- Presence of noise and signal attenuation.
- Non ideality of zobel network on output leads to cracking and hissing sounds.
- Circuit is mechanically unstable due to congestion in breadboard and unstable mechanical structure of coils

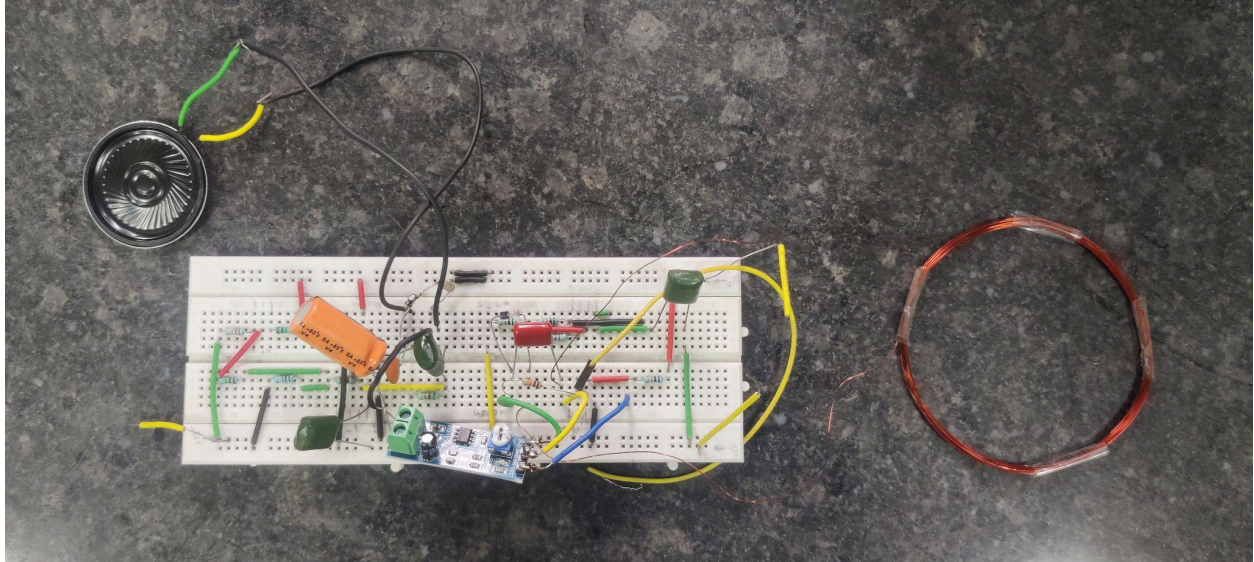
Future improvements

- Designing a structure to keep transmitting and receiver coils stable mechanically.
- Solving problem of breadboard instability and congestion by soldering entire circuit together properly.
- Designing a proper Zobel network for the output speaker.
- To enter domain of radio communications proper antenna design is required which is a lot easier at higher frequencies.
- Using a high frequency amplifier after modulator to amplify the transmitted wave hence increasing output power.
- Wireless image transmission can be done from the same setup once the speaker circuit is stable using Slow static television.

Circuit/Setup pictures



Transmitter circuit



Receiver circuit

Note :- There is a other circuit also mounted on the receiver breadboard due to unavailability of empty breadboard

Novelty and Possible applications

The present work demonstrates a proof of concept and is limited to a short range of 4-5cm, but with enhanced and improved versions with electromagnetic waves dominating the communication, with extended range and better audio quality it has several applications in real world

Short range control systems and wireless audio transmission

With extended transmission range, better output amplification and improved reliability, future versions of this system may be applied in short-range wireless control systems such as:

- Remote switching systems
- Wireless doorbell systems
- Device triggering and signaling circuits

Such systems can benefit from localized communication, cheap manufacturing and low power usage. Also future versions can provide higher-quality wireless audio transmission for short-range communication applications.